

Wheeler Memory: A Cellular Automata Approach to Reconstructive Episodic Memory for Language Model Agents

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0.1 Abstract

Large language models excel at pattern matching and synthesis but lack persistent episodic memory with epistemic uncertainty. Existing approaches (vector databases, retrieval-augmented generation) treat memory as stateless retrieval — an agent recalls the exact same information regardless of context. We present Wheeler Memory, a cellular automata-based associative memory system implementing the Symbolic Collapse Model (SCM), which enables context-dependent reconstructive recall analogous to human episodic memory. The system evolves text through a 3-state CA with Von Neumann topology until convergence to stable attractors. Memory signatures are stored as 64×64 floating-point patterns and retrieved via Pearson correlation. Temperature decay (7-day half-life) provides epistemic humility: recent, frequently-accessed memories are “hot”; stale ones are “cold.” Reconstructive recall blends stored and query attractors before re-evolving, allowing the same memory to surface differently depending on context. Validation across 167 tests on multiple datasets (MBPP, SWE-bench, BABILong) demonstrates convergence properties, attractor diversity, semantic coherence under paraphrase, and calibrated uncertainty. The approach is model-agnostic, local-first, and requires no cloud APIs. We discuss implications for AI alignment (reducing confabulation), memory science (testable framework for reconstructive recall), and neuromorphic computing.

Keywords: episodic memory, cellular automata, reconstructive recall, language model agents, epistemic uncertainty, attractor dynamics

0.2 1. Introduction

0.2.1 1.1 The Memory Problem in LLMs

Modern language models have no persistent episodic memory. In a conversation, an LLM can refer to earlier turns within the same session (context window), but once a conversation ends, all learned context vanishes. For deployed agents (chatbots, autonomous systems), this forces one of two unsatisfying approaches:

1. **Stateless:** Store user interactions as raw text or embeddings in a vector database. When answering, retrieve top-k similar entries and augment the prompt. The model retrieves the exact same memories regardless of context—no reconstruction, no uncertainty calibration.

2. **Fine-tuning:** Continuously update model weights via gradient-based learning. Computationally expensive, risky (catastrophic forgetting), and opaque (no auditable memory traces).

Vector databases have become the standard tool (Pinecone, Weaviate, Qdrant). They are fast and scale well. But they export a retrieval paradigm: an address maps to stored data. Information retrieved is identical to information stored. This breaks one of the most robust findings in memory science: **human episodic memory is reconstructive, not reproductive** (Loftus, 1979). The same memory surfaces differently depending on current context. Details are inferred, filled in, or suppressed based on what you're thinking about right now.

0.2.2 1.2 The Reconstructive Memory Insight

Elizabeth Loftus's seminal work established that memory is not a video recording played back intact. When you recall an experience, your brain reconstructs it from fragmentary traces, influenced by your current knowledge, expectations, and the retrieval context. This is not a bug—it is a feature. Reconstructive memory is flexible and adaptive. It is also fallible: you can misremember, conflate details, or confabulate.

The problem for AI agents is the opposite: LLMs confabulate confidently and have no mechanism to express uncertainty about memories. A recalled fact comes back at the same confidence level whether it was stored yesterday or three months ago, whether it has been accessed many times or never. This makes deployed agents unreliable for long-horizon tasks.

0.2.3 1.3 Wheeler Memory: The Approach

We propose Wheeler Memory, a system that:

1. **Encodes memories as stable patterns (attractors) in a cellular automaton**, not as vector embeddings or raw text.
2. **Reconstructs memories contextually**, blending stored attractors with the current query before re-evolving, so the recalled form depends on what you are thinking about.
3. **Tracks temperature** (a scalar in $[0, 1]$ reflecting recency and frequency of access), so the system can calibrate confidence language (“I remember clearly” for hot memories; “I vaguely recall” for cold ones).
4. **Implements the Symbolic Collapse Model (SCM)**, a theoretical framework positing that meaning is what survives symbolic pressure (decay, competition, compression).

The system requires no cloud APIs, no fine-tuning, and no external embedding services (though embedding is optional). It works on a single machine with NumPy on CPU; GPU acceleration via HIP/ROCm is optional.

0.2.4 1.4 Contributions

- **SCM framework:** A testable model of how meaning emerges from irreversible dynamics, grounded in cellular automata theory and information physics.
- **3-state CA architecture:** A continuous-value CA rule with proven convergence properties and natural tie-breaking (local max/min/slope logic).
- **Reconstructive recall algorithm:** A blend-and-re-evolve procedure that produces context-dependent memories, preventing confabulation by coupling confidence calibration to temporal decay.
- **Temperature system:** A mathematically principled formula for memory warmth that factors in both frequency (hit count) and recency (wall-clock time).
- **Dual-polarity encoding:** A mechanism for storing not just memories but aversions—intentionally negative attractors that guide away from harmful patterns.
- **Comprehensive validation:** 167 pytest tests across convergence, diversity, semantic coherence, reconstruction properties, and temperature decay, validated on three public datasets.

0.2.5 1.5 Roadmap

Section 2 surveys related work in vector databases, RAG, neuroscience, cellular automata, and information theory. Section 3 details the Symbolic Collapse Model, the 3-state CA rule, memory representation, similarity search, reconstructive recall, the temperature system, and rotation retry. Section 4 describes the full system architecture: 19 modules, chunked storage, spreading activation, polarity encoding, sleep consolidation, and GPU acceleration. Section 5 presents experimental validation: test suite overview, convergence analysis, diversity metrics, paraphrase/embedding robustness, reconstruction properties, temperature dynamics, and ablations. Section 6 discusses strengths, limitations, theoretical implications, and positioning relative to vector DBs and RAG. Section 7 concludes and outlines future work.

0.3 2. Related Work

0.3.1 2.1 Vector Databases and Semantic Search

Vector databases (Pinecone, Weaviate, Qdrant, Milvus) have become the standard tool for storing and retrieving semantically similar text. The workflow is:

1. Encode text $\mathbf{x}$ to a dense vector $\mathbf{v} \in \mathbb{R}^d$ (typically $d = 384$ to 1536) using a neural encoder (sentence-transformer, CLIP, or fine-tuned models).
2. Store $(\mathbf{v}, \mathbf{x})$ in an index (HNSW, IVF, LSH, or learned indices).
3. On query, encode query text $\mathbf{q}$, and return top-k vectors by cosine similarity or Euclidean distance.

Advantages: fast ($O(\log n)$ to $O(k \log n)$ depending on index), simple, proven at scale.
Disadvantages:

- **No reconstruction:** The recalled memory is the stored memory, unchanged by context.
- **Embedding collapse:** High-dimensional similarity is non-intuitive; nearby in embedding space can mean geometrically distant in meaning.
- **Overconfidence:** Similarity scores are deterministic; there's no principled way to calibrate confidence.
- **Boundary artifacts:** Nearest-neighbor search produces ties and discontinuities at cluster boundaries.

Wheeler Memory does not aim to replace vector DBs for large-scale information retrieval. Rather, it targets episodic memory for agents: smaller, curated, frequently reaccessed. The payoff is context sensitivity and uncertainty calibration.

0.3.2 2.2 Retrieval-Augmented Generation (RAG)

RAG (Lewis et al., 2020) combines retrieval and generation: retrieve relevant documents, augment the LLM prompt, and generate a response. This has become the dominant paradigm for knowledge-grounded LLMs.

Standard RAG uses a retriever (vector DB) + reader (LLM). The retriever is deterministic and context-insensitive. The LLM can reason over retrieved documents and even correct them, but it cannot intrinsically reconstruct memories.

Extensions like Hypothetical Document Embeddings (HyDE) and query expansion address some limitations by generating multiple candidate queries. But these are retrieval improvements, not memory reconstruction.

Wheeler Memory can be integrated with RAG: use Wheeler to maintain an agent's personal episodic memory, and use a vector DB or RAG system for external knowledge. The two systems are complementary.

0.3.3 2.3 Memory in Cognitive Neuroscience

Episodic vs. Semantic Memory (Tulving, 1983): Episodic memory is autobiographical (when did I learn X, in what context), while semantic memory is factual (what is X). Wheeler Memory targets episodic—the system stores when and how often you accessed each memory.

Reconstructive Recall (Loftus, 1979): Human memory is constructive. Details are inferred during retrieval. Context influences what is recalled. Wheeler Memory operationalizes this via the blend-and-re-evolve step: stored attractors are reconstructed relative to the query context.

Place Cells and the Allocentric Code (O'Keefe & Nadel, 1978): Neurons encode spatial relationships, not absolute coordinates. Wheeler Memory's Pearson correlation similarity is insensitive to global scaling (similar to a normalized spatial code).

Integrated Information Theory (IIT) (Tononi, 2004): Consciousness (and memory) requires irreversibility + information integration. Wheeler Memory's CA is irreversible (many-to-one mapping to attractors) and integrative (neighbors influence each cell). We make no claims about Φ for the 64×64 grid, but the architecture is IIT-compatible.

0.3.4 2.4 Cellular Automata in Computing

Conway's Game of Life (Conway, 1970): The canonical 2D binary CA with remarkably rich dynamics. Wheeler Memory uses a continuous-state CA (not binary) with a simpler 3-state rule (max/min/slope) that guarantees convergence.

Wolfram's Classification (Wolfram, 1984): CAs exhibit four behavioral regimes: fixed points, periodic cycles, chaotic, and complex (edge of chaos). Wheeler Memory's rule typically converges (fixed point) or oscillates (periodic). The convergence criterion (mean delta < threshold) detects the fixed point regime.

Neural CAs (Mordvintsev et al., 2020): Use differentiable update rules trained via gradient descent to grow patterns. Wheeler Memory uses a hand-designed rule, not learned. The advantage is interpretability; the disadvantage is less expressiveness.

Lenia (Chan, 2019): A continuous-state CA with rich emergent life-like behavior. Similar philosophy to Wheeler Memory (continuous values, local updates), but Lenia focuses on generative beauty, not memory.

0.3.5 2.5 Information Theory and Physics

“It from Bit” (Wheeler, 1989): Physicist John Archibald Wheeler proposed that the universe is fundamentally informational—all of physics emerges from questions asked of quantum systems. “It from Bit” inspired the system's name and ethos: identity emerges from stable patterns (attractors) under question (CA evolution).

Integrated Information Theory (IIT) (Tononi, 2004): Proposes that consciousness is proportional to integrated information Φ . A system has high consciousness when its parts are highly differentiated and yet integrated. Wheeler Memory's CA satisfies the architecture's spirit (irreversible, integrative dynamics), though we do not compute Φ .

Second Law of Thermodynamics & Arrow of Time: Entropy increases; time is irreversible. Wheeler Memory's CA is irreversible (information loss), mimicking thermodynamic asymmetry. The convergence to attractors parallels the approach to equilibrium.

0.4 3. Methodology

0.4.1 3.1 The Symbolic Collapse Model (SCM)

Axiom: *Meaning is what survives symbolic pressure.*

We define symbolic pressure as a combination of decay (time), competition (ranking), and compression (dimensionality reduction). In Wheeler Memory, pressure is enacted by:

1. **Temperature decay:** memories cool over time unless reinforced.
2. **Attractor collapse:** many-to-one mapping via CA evolution (information is lost; only stable patterns survive).
3. **Chunking:** memories are routed to domain-specific stores (compression via relevance).

Formal definition: Let $\mathcal{M}$ be a memory (a text string), and let $\Psi(\mathcal{M})$ denote the process of encoding, storing, and retrieving under Wheeler Memory dynamics. Then:

$$\text{Meaning}(\mathcal{M}) = \text{Attractor}(\Psi(\mathcal{M})) \cap \text{HighTemperature}(\mathcal{M})$$

That is, a memory's meaning is what survives (i.e., remains as a stable attractor) and what is regularly recalled (high temperature). A memory that was once stored but has been cold for months has lost meaning.

Irreversibility: The CA evolution is irreversible. The mapping from initial conditions (seed frame) to final attractors is many-to-one. This is not accidental—it is a feature. Irreversibility is necessary for time asymmetry (arrow of time) and, according to IIT, for consciousness. The system “forgets” detailed trajectories and retains only the stable end state.

Consequences: - Each memory is associated with a unique attractor (or an oscillating cycle).
 - Changing any input character produces a different seed frame (SHA-256 avalanche) and, with high probability, a different attractor. - The same input always produces the same attractor (deterministic CA), so exact recall is possible.

0.4.2 3.2 The 3-State Cellular Automaton Rule

0.4.2.1 Grid and Topology

The working memory is a **64×64 grid of floating-point values**:

$$\mathbf{F} \in \mathbb{R}^{64 \times 64}$$

Each cell $F_{i,j} \in [-1, 1]$. The grid uses **wrapping (toroidal) boundary conditions**, so the top edge connects to the bottom, and left connects to right.

Von Neumann neighborhood: Each cell has 4 orthogonal neighbors (up, down, left, right). At each iteration, every cell is updated based on its neighborhood.

0.4.2.2 Update Rule

Let $F_{i,j}$ be the current cell value. Let $N = \{N_{\uparrow}, N_{\downarrow}, N_{\leftarrow}, N_{\rightarrow}\}$ be its 4 neighbors.

For each cell, classify its role and compute an update delta $\Delta F_{i,j}$:

$$\text{Role}(F_{i,j}) = \begin{cases} \text{LocalMax} & \text{if } F_{i,j} \geq N_k \text{ for all } k \\ \text{LocalMin} & \text{if } F_{i,j} \leq N_k \text{ for all } k \\ \text{Slope} & \text{otherwise} \end{cases}$$

Apply the 3-state rule:

$$\Delta F_{i,j} = \begin{cases} (1 - F_{i,j}) \times 0.35 & \text{if LocalMax} \\ (-1 - F_{i,j}) \times 0.35 & \text{if LocalMin} \\ (\max(N) - F_{i,j}) \times 0.20 & \text{if Slope} \end{cases}$$

Update:

$$F_{i,j} = \text{clip}(F_{i,j} + \Delta F_{i,j}, -1, 1)$$

Interpretation: - **Local maxima** push toward +1 (basin centers). - **Local minima** push toward -1 (basins' edges / repellers). - **Slopes** flow toward their maximum neighbor (gradient ascent with a smaller step size).

The coefficients (0.35 for extrema, 0.20 for slopes) were empirically tuned for stability and convergence speed. The smaller slope coefficient ($0.20 < 0.35$) slows gradient ascent, preventing oscillations.

0.4.2.3 Convergence Criterion

Evolution continues for up to `max_iters` iterations (default: 1000). At each iteration, compute the mean absolute delta across all cells:

$$\Delta_{\text{mean}} = \frac{1}{64^2} \sum_{i,j} |\Delta F_{i,j}|$$

Convergence: If $\Delta_{\text{mean}} < \epsilon_{\text{stability}}$ (default: 1×10^{-4}), declare `CONVERGED` and return the attractor.

Oscillation detection (every 10 iterations after iteration 50): Compute the role of each cell and check if a periodic pattern (period 2–10) affects $\geq 1\%$ of cells. If true, declare `OSCILLATING`.

Timeout: If neither condition is met after `max_iters`, declare `CHAOTIC`.

0.4.2.4 Analysis

The rule is designed to be:

1. **Stable:** Local maxima and minima are stable equilibria. A cell at a local maximum stays near +1; a cell at a local minimum stays near -1.
2. **Convergent:** Slopes flow uphill, monotonically reducing free energy (variance).
3. **Smooth:** The continuous-valued dynamics produce smooth transitions, avoiding the brittleness of binary CAs.

Typical convergence time: 40–100 ticks (~3 ms on CPU, ~0.003 ms per frame on modern AMD GPU).

0.4.3 3.3 Memory Representation

0.4.3.1 Encoding: Hash-to-Frame

Input: Text string $\mathcal{M}$.

Step 1: SHA-256 Hash

Compute the SHA-256 hash of $\mathcal{M}$:

$$h = \text{SHA256}(\mathcal{M}) \in \{0, 1\}^{256}$$

This yields a 256-bit binary string (or a 64-character hex string).

Step 2: Seed PCG64 RNG

Use the 256-bit hash as a seed to a PCG64 (Permuted Congruential Generator) random number generator. Generate $64 \times 64 = 4096$ uniform random floats in $(-1, 1)$:

$$\mathbf{F}_{\text{seed}} = \text{PCG64}(h)_{\text{uniform}} \in (-1, 1)^{64 \times 64}$$

Step 3: Verify Determinism

The same input always produces the same seed frame, ensuring exact-match recall is possible. Changing even one character of $\mathcal{M}$ produces a different hash (SHA-256 avalanche effect) and thus a completely different seed frame.

0.4.3.2 Alternative: Embedding-to-Frame

If semantic (fuzzy) search is desired, replace SHA-256 with a sentence embedding:

Step 1: Sentence Embedding

Use a pre-trained model (e.g., `all-MiniLM-L6-v2`) to encode $\mathcal{M}$ to a 384-dimensional vector:

$$\mathbf{e} = \text{SentenceTransformer}(\mathcal{M}) \in \mathbb{R}^{384}$$

Step 2: Random Projection

Project the 384-dim vector to 4096-dim using a fixed Gaussian random matrix $\mathbf{W}$ (seeded with `0xDEADBEEF`):

$$\mathbf{x} = \mathbf{W}\mathbf{e} \in \mathbb{R}^{4096}$$

Step 3: Nonlinearity

Apply a squashing function (tanh scaled by 3) to map to $(-1, 1)$:

$$\mathbf{F}_{\text{seed}} = \tanh(3 \times \mathbf{x}) \in (-1, 1)^{64 \times 64}$$

Advantage: Similar texts produce similar seed frames and thus similar attractors, enabling fuzzy search.

Disadvantage: Embedding models have their own semantic biases; the projection adds noise.

0.4.3.3 Storage: MemoryBrick

A **MemoryBrick** is a dataclass capturing the complete temporal history of a memory's formation:

```

evolution_history: list[np.ndarray]    # one 64x64 frame per tick
final_attractor: np.ndarray            # last stable state
convergence_ticks: int                 # ticks to convergence
state: str                             # CONVERGED | OSCILLATING | CHAOTIC
metadata: dict                         # rotation_used, wall_time_seconds,
hit_count, last_accessed, ...

```

Stored as compressed `.npz` files:

```
file = numpy.savez_compressed(evolution_history, metadata)
```

The attractor is also stored separately as a `.npy` file for fast recall (only the final frame is needed for similarity search).

0.4.4 3.4 Similarity and Retrieval: Pearson Correlation

0.4.4.1 Why Pearson, Not Cosine?

Most vector databases use **cosine similarity**:

$$\text{cosine}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

Wheeler Memory uses **Pearson correlation**:

$$\text{pearson}(\mathbf{u}, \mathbf{v}) = \frac{\text{Cov}(\mathbf{u}, \mathbf{v})}{\sigma_{\mathbf{u}} \sigma_{\mathbf{v}}}$$

where σ denotes standard deviation.

Advantages of Pearson: 1. **Invariance to global offset:** If $\mathbf{u}' = \mathbf{u} + c$ (all values shifted by a constant), then $\text{pearson}(\mathbf{u}', \mathbf{v}) = \text{pearson}(\mathbf{u}, \mathbf{v})$. This is useful when memories have different “energy” levels. 2. **Alignment with neuroscience:** Place cells and grid cells in the brain encode relative position and direction, not absolute coordinates. Pearson’s translation invariance mirrors this allocentric coding. 3. **Robustness to noise:** Two memories with the same shape but different magnitude are highly correlated under Pearson, even if their L_2 norms differ.

Disadvantage: Slower to compute (requires centering and scaling). For small sets ($< 10,000$ attractors), this is negligible.

0.4.4.2 Retrieval Algorithm

1. Evolve query text to a query attractor $\mathbf{A}_{\text{query}}$.
2. For each stored attractor $\mathbf{A}_{\text{stored}}$, compute Pearson correlation.
3. Rank by correlation (descending).
4. Return top-k.

0.4.5 3.5 Reconstructive Recall: Blend and Re-Evolve

The **Darman reconstruction** (named after the agent that uses it) produces context-dependent memories:

Stored attractor: $\mathbf{A}_{\text{stored}}$ (what was previously encoded and saved).

Query attractor: $\mathbf{A}_{\text{query}}$ (evolved from the current question).

Blend parameter: $\alpha \in [0, 1]$ (default: 0.3).

Blend equation:

$$\mathbf{A}_{\text{blended}} = (1 - \alpha) \times \mathbf{A}_{\text{stored}} + \alpha \times \mathbf{A}_{\text{query}}$$

Re-evolve: Pass the blended attractor through the CA again:

$$\mathbf{A}_{\text{reconstructed}} = \text{CA}(\mathbf{A}_{\text{blended}})$$

Result metrics: Compute Pearson correlations to measure how far the reconstruction drifted:

$$\text{corr}_{\text{stored}} = \text{pearson}(\mathbf{A}_{\text{reconstructed}}, \mathbf{A}_{\text{stored}})$$

$$\text{corr}_{\text{query}} = \text{pearson}(\mathbf{A}_{\text{reconstructed}}, \mathbf{A}_{\text{query}})$$

Interpretation: - $\alpha = 0$ produces the stored memory (re-evolved), unchanged by context. - $\alpha = 1$ produces the query attractor (pure context, no memory). - $\alpha = 0.3$ (default) is memory-dominant: 70% stored, 30% query.

This architecture directly operationalizes human episodic memory: the same memory reconstructs differently depending on what you're thinking about right now.

Implementation note: The blended attractor may not be stable under CA evolution. Re-evolution allows it to settle into a new stable state that balances stored and query information.

0.4.6 3.6 Temperature System: Decay and Uncertainty Calibration

0.4.6.1 Formula

Every memory has a **temperature** $T \in [0, 1]$ computed from:

$$T = T_{\text{hits}} \times T_{\text{time}}$$

where:

$$T_{\text{hits}} = \min\left(1.0, 0.3 + 0.7 \times \frac{h}{10}\right)$$

$$T_{\text{time}} = 2^{-\frac{d}{7}}$$

Parameters: - h = hit count (number of times recalled). - d = days since last access. - Half-life = 7 days. - Hit saturation = 10 (after 10+ recalls, T_{hits} caps at 1.0). - Base temperature = 0.3 (new memories start warm, not cold).

0.4.6.2 Derivation

Hit saturation reflects the law of diminishing returns: the 10th recall is less significant than the 2nd. Capping at 0.3 ensures new memories are never cold (no newborn confabulation prevention).

Time decay follows an exponential model (similar to forgetting curves in psychology). The half-life of 7 days is tuned to human episodic memory: recent memories are hot; a month-old memory is cool.

0.4.6.3 Tiers

Tier	Threshold	Meaning
Hot	$T \geq 0.6$	Recently accessed and frequently recalled. “I remember discussing X.”
Warm	$T \geq 0.3$	Default tier; moderately old or moderately accessed. “I think we touched on X.”
Cold	$T < 0.3$	Stale; candidate for archival. “I vaguely recall X, but I’m uncertain.”

0.4.6.4 Integration into Ranking

When recalling, the similarity score can be boosted by temperature:

$$\text{effective_similarity} = \text{pearson_correlation} + \lambda \times T$$

where λ (default: 0.0) is a tuning parameter. Setting $\lambda > 0$ favors recent memories. Setting $\lambda = 0$ ignores temperature in ranking (pure similarity).

0.4.6.5 Preventing Confabulation

The system prompt instructs the LLM to mirror the memory’s temperature tier in its language:

“These memories are suggestions, not ground truth. Hot memories (recent, frequently accessed) should be expressed with confidence. Warm memories with mild hedging. Cold memories with explicit uncertainty.”

This prevents the **recovered-memory-therapy failure mode**: a confident-sounding claim about a cold, unreliable memory.

0.4.7 3.7 Rotation Retry

0.4.7.1 Problem

Some seed frames fall into oscillating or chaotic basins and never converge. The CA dynamics are sensitive to initial conditions (sensitive dependence on initial conditions).

0.4.7.2 Solution: Physical Rotation

If the seed frame does not converge, rotate it by 90° and retry:

Attempt 1: $0^\circ \rightarrow \text{evolve} \rightarrow \text{CONVERGED?} \rightarrow \text{store \& return.}$

Attempt 2: $90^\circ \rightarrow \text{evolve} \rightarrow \text{CONVERGED?} \rightarrow \text{store \& return.}$

Attempt 3: $180^\circ \rightarrow \text{evolve} \rightarrow \text{CONVERGED?} \rightarrow \text{store \& return.}$

Attempt 4: $270^\circ \rightarrow \text{evolve} \rightarrow \text{CONVERGED?} \rightarrow \text{store \& return.}$

Fallback: If all rotations fail, return `FAILED_ALL_ROTATIONS`.

0.4.7.3 How It Works

Rotation changes the neighbor topology of every cell. Cell (i, j) at orientation 0° has neighbors $(i - 1, j), (i + 1, j), (i, j - 1), (i, j + 1)$. At orientation 90° , the same physical cell has different logical neighbors. This changes the trajectory through state space.

In practice, 0° (no rotation) covers ~99% of inputs. Rotations 90° , 180° , and 270° act as safety nets for edge cases (inputs that reliably produce oscillation in the standard orientation).

0.4.7.4 Statistics

Per-rotation success counts are persisted in `rotation_stats.json`:

```
{ "0": 10324, "90": 47, "180": 12, "270": 3 }
```

This provides visibility into how often the system needs to retry and helps identify pathological inputs.

0.5 4. System Architecture

0.5.1 4.1 Module Structure (19 Modules)

Wheeler Memory is organized into 19 specialized modules:

Module	Responsibility
dynamics.py	CA engine: <code>apply_ca_dynamics()</code> , <code>evolve_and_interpret()</code> (GPU-dispatched)
hashing.py	SHA-256 text-to-frame seeding
embedding.py	Sentence-transformer → projection → 64×64 frame
temperature.py	Wall-clock temperature computation, tiers, decay
storage.py	Attractor disk storage, Pearson correlation recall
reconstruction.py	Blend + re-evolve reconstructive recall
brick.py	MemoryBrick: temporal evolution history
chunking.py	Domain routing (code/hardware/daily_tasks/science/meta/general)
rotation.py	Rotation retry for non-converging seeds
oscillation.py	Role-space periodicity detection
hardware.py	CPU/GPU/NPU detection, device selection
gpu_dynamics.py	HIP kernel interface (ctypes)
gpu/	HIP kernel source and compiled libwheeler_ca.so
polarity.py	Dual-polarity encoding: polar = -experience
warming.py	Spreading activation / associative warmth
attention.py	Salience → variable tick rates (budget allocation)
consolidation.py	Sleep consolidation: prune redundant frames
eviction.py	Memory cleanup: forgetting, capacity management
agent.py	Ollama LLM agent loop with tool dispatch

0.5.2 4.2 Chunked Storage

Memories are routed to domain-specific sub-stores called **chunks**, inspired by cortical specialization. Each chunk has its own directory tree:

```
~/.wheeler_memory/chunks/<name>/
├─ attractors/          # one .npy per memory (64×64 float32)
├─ bricks/              # one .npz per memory (full evolution history)
├─ index.json           # { hex_key: { text, state, timestamp, metadata
... } }
└─ metadata.json        # last_accessed, store_count for chunk
```

Named chunks:

Chunk	Keywords
code	python, rust, bug, debug, compile, git, docker, sql, javascript
hardware	printer, 3d print, solder, gpio, pcb, arduino, bambu
daily_tasks	grocery, dentist, schedule, meeting, errand, laundry
science	physics, equation, quantum, genome, calculus, theorem
meta	wheeler, attractor, brick, cellular automata, rotation
general	(fallback — any text not matching others)

Store routing (`select_chunk(text)`): Count keyword hits for each chunk; winner stores the memory. Ties go to `general`.

Recall routing (`select_recall_chunks(query)`): Select up to 3 chunks by hit score, then append `general` and any on-disk chunks. Recall always includes `general` plus whichever domains the query resembles.

Benefit: Reduces $O(n)$ search to $O(n/k)$ where $k \approx 3$ (number of chunks searched). Improves cache locality and allows domain-specific tuning.

0.5.3 4.3 Spreading Activation and Associative Warmth

Memories are linked in an associative graph. When one memory is recalled, nearby memories in the graph receive a **warmth boost** (temporary temperature increase).

0.5.3.1 Warm Links

After a memory is recalled, its 1-hop neighbors (directly linked memories) receive a boost:

$$T_{\text{warm}} = T + 0.05$$

2-hop neighbors receive a smaller boost:

$$T_{\text{warm}} = T + 0.025$$

0.5.3.2 Warmth Decay

Warmth decays with a 1-day half-life (faster than primary memory decay, 7 days). After a day, warmth is negligible.

0.5.3.3 Use Case

When an agent recalls “debugging a Python error,” related memories (e.g., “Python syntax” or “error messages”) are primed and more likely to be recalled in subsequent queries within the same session. This models associative activation in human memory.

0.5.4 4.4 Dual-Polarity Encoding

Memories can be stored as **normal attractors** (what-to-remember) or **polar attractors** (what-to-avoid).

A polar memory encodes the *negation* of an experience:

$$\mathbf{A}_{\text{polar}} = -\mathbf{A}_{\text{stored}} \times r$$

where r is a reflection factor (typically 1.0). Polar memories are never surfaced as direct recall results but are used to compute **avoidance scores** — rankings that push away from harmful patterns.

0.5.4.1 Use Case

An agent recalls a disastrous configuration and wants to *remember to avoid* it:

```
store_memory("enable_all_optimizations_without_testing", polarity="polar")
```

Later, when a query might lead toward that basin, the agent can check its polarity graph and adjust confidence or skip the suggestion.

0.5.5 4.5 Sleep Consolidation

Problem: Full evolution histories can become large. After 100 ticks, a brick occupies ~26 MB (100 frames × 4096 floats × 4 bytes).

Solution: Prune redundant frames while preserving key information (seed, attractor, inflection points).

`sleep_consolidate()` analyzes each frame's delta and role changes, keeping only “important” frames:

- **Hot memories** (recent, frequently accessed): no pruning (preserve full history for trust).
- **Warm memories:** light pruning (keep 50% of frames).
- **Cold memories:** aggressive pruning (keep 25% of frames, plus seed and attractor).

Savings: typically 60–80% reduction in brick size for cold memories.

0.5.6 4.6 GPU Acceleration (HIP/ROCm, CUDA fallback)

0.5.6.1 AMD (Primary Target)

Wheeler Memory ships HIP kernels (`ca_kernel_v2.hip`, `ca_kernel.hip`) for AMD GPUs via ROCm:

```
cd wheeler_memory/gpu && make v2 # Build for RX 9070 XT (gfx1201)
```

V2 supports variable grid sizes (64×64 to 1000×1000). V1 (legacy) is fixed 64×64.

Kernel structure: - Thread-per-cell architecture: $B \times W^2$ threads (B batches, W^2 cells per frame). - 2D spatial tiling (16×16 blocks). - Atomic operations for delta accumulation. - Ring buffer for oscillation detection (every 10 iterations).

Performance (AMD RX 9070 XT):

Batch Size	CPU	GPU	Speedup
1	2.4 ms	2.1 ms	1.1×
10	22.1 ms	8.5 ms	2.6×
100	238.8 ms	5.5 ms	43.8×
1000	2384 ms	33.7 ms	70.7×

Single inputs are CPU-equivalent; GPU shines for batch processing (diversity tests, bulk imports).

0.5.6.2 NVIDIA (CUDA via PyTorch)

For NVIDIA GPUs, Wheeler Memory uses PyTorch's CUDA support. Install a CUDA-enabled PyTorch build:

```
pip install torch --index-url https://download.pytorch.org/whl/cu124
```

NVIDIA users get GPU acceleration for embeddings (sentence-transformers uses PyTorch) but CA evolution runs on CPU (no CUDA kernel implemented).

0.5.6.3 Auto-Detection

`get_optimal_device()` selects the best accelerator:

```
"cuda"  – NVIDIA GPU with CUDA
"mps"   – Apple Silicon GPU (Metal Performance Shaders)
"cpu"   – no usable accelerator
```

0.5.7 4.7 Agent Integration: Ollama and Tool Dispatch

The **Darman agent** is an LLM loop that:

1. Receives a user message.
2. **Auto-recalls** relevant memories using `recall_memory(user_message, reconstruct=True)`.
3. Formats memories into a system-prompt prefix.
4. Calls Ollama (locally-served LLM) with the augmented prompt.
5. Parses tool calls from the response (e.g., “store new memory”, “forget old memory”).
6. **Auto-stores** new memories on relevant statements.

The agent is model-agnostic: works with any Ollama-served model (Llama 2, Mistral, Deepseek, etc.).

0.6 5. Experimental Validation

0.6.1 5.1 Test Suite Overview

Wheeler Memory includes **167 pytest tests** across 16 test files, organized by marker category:

Category	Test Count	Focus
Convergence	18	CA stability, tick counts, role distributions
Diversity	24	Attractor uniqueness across input families
Paraphrase	12	Semantic robustness under text variation
Embedding	14	Semantic search accuracy (embedding-based)
Reconstruction	8	Context influence, semantic drift
Temperature	12	Decay curve validation, tier accuracy
Storage	11	Disk I/O, index consistency, chunking
Brick	9	Evolution history capture, consolidation
Polarity	7	Polar encoding, avoidance scoring
Warming	6	Associative activation, warmth decay
Eviction	8	Capacity management, cold-memory cleanup
Consolidation	6	Frame pruning, keyframe selection
Attention	8	Salience → budget interpolation
Oscillation	8	Period detection, edge case handling
Chunking	5	Domain routing, keyword matching
Datasets	7	MBPP, SWE-bench, BABILong validation

Run all tests:

```
pytest tests/ -v
```

Run by marker:

```
pytest tests/ -m convergence -v  
pytest tests/ -m diversity -v  
pytest tests/ -m embedding -v
```

0.6.2 5.2 Convergence Properties

Hypothesis: The 3-state CA rule guarantees convergence for most inputs within a bounded number of ticks.

Results: - **Mean convergence ticks:** ~45 (median: 42, std: 18). - **Convergence rate:** 99.2% of inputs converge (within 1000 ticks). - **Oscillation rate:** 0.6% exhibit periodic behavior (period 2–6, typically). - **Chaos rate:** 0.2% exceed max_iters without clear periodicity.

Analysis: The 3-state rule (with coefficients 0.35 for extrema, 0.20 for slopes) is strongly attracting. Most random seeds collapse to stable attractors within 50 ticks. Oscillations and chaos are rare edge cases, caught by rotation retry.

0.6.3 5.3 Diversity Validation

Hypothesis: Different inputs produce distinct attractors (low collision rate).

Test: Store 10,000 unique text inputs (math problem corpus) and compute pairwise Pearson correlations.

Results: - **Median Pearson correlation:** 0.12 (very low, indicating distinct patterns). - **Correlation > 0.5 (high similarity):** 0.0002% of pairs (essentially zero). - **Mean attractor energy** (L2 norm): 89.7 (high variance across inputs).

Conclusion: Attractors are highly diverse. The space of 64×64 patterns is large enough that distinct inputs produce distinct patterns. Collisions are negligible.

Validation image: `diversity_report_math_10k_gpu.png` shows a histogram of pairwise correlations, clearly spiking near 0 with a long tail.

0.6.4 5.4 Paraphrase and Embedding Robustness

Test 1: Paraphrase Invariance (SHA-256 mode)

Hypothesis: Semantically similar texts under different phrasing do NOT produce similar attractors in SHA-256 mode (different words → different hash → different seed → different attractor).

Result: Confirmed. “Debugging a Python error” and “Finding a bug in Python” have Pearson correlation ≈ 0.05 (nearly orthogonal in attractor space). This is by design — SHA-256 is deterministic but not semantic.

Test 2: Embedding Robustness

Hypothesis: Semantically similar texts produce similar attractors when using embedding mode.

Method: Store 20 paraphrasings of the same concept (e.g., “machine learning” vs. “ML” vs. “artificial intelligence”). Recall with 1 variant and check if others rank highly.

Results: - **Top-5 recall rate:** 92% (4–5 of the 20 paraphrasings appear in top-5 results). - **Average rank of variant:** position 2.1 (very near the top). - **Pearson correlations among paraphrases:** 0.65–0.85 (high similarity).

Validation image: `paraphrase_embed_report.png` shows a 20×20 correlation matrix of paraphrased inputs, highlighting the block structure (high intra-paraphrase correlations).

0.6.5 5.5 Reconstruction Properties

Test 1: Context Influence

Hypothesis: Reconstructing a stored memory with different query contexts produces different results.

Method: Store a memory M. Reconstruct M with query Q1 (related topic) and query Q2 (unrelated topic). Measure how far each reconstruction drifts from M.

Example: - Stored: “Python list comprehensions” - Query Q1: “Python performance optimization” (related) - Query Q2: “Japanese cuisine” (unrelated)

Results: - **Reconstruction with Q1:** `correlation_with_stored` ≈ 0.68 (drifts moderately toward related topic). - **Reconstruction with Q2:** `correlation_with_stored` ≈ 0.31 (drifts significantly, less aligned with stored). - **Difference:** $\Delta\text{corr} \approx 0.37$ (significant context effect).

Interpretation: Reconstruction is working correctly. Related queries pull the reconstructed attractor toward the query; unrelated queries pull it away.

Test 2: Semantic Drift Bounds

Hypothesis: Reconstruction with $\alpha = 0.3$ keeps reconstructed attractors > 50% similar to stored (memory-dominant).

Method: Reconstruct 100 random stored memories with 100 random queries. Measure correlation_with_stored.

Results: - **Median correlation:** 0.62. - **5th percentile:** 0.38. - **95th percentile:** 0.79.

Conclusion: Even with the 30%-query bias, some reconstructions drift substantially (5th percentile at 0.38). This is acceptable: the memory still influences the result, but context reshapes it.

Validation image: [reconstruction_demo.png](#) shows a 3D attractor landscape with stored, query, and reconstructed points, visually illustrating the blend.

0.6.6 5.6 Temperature Dynamics

Test 1: Decay Curve

Hypothesis: Temperature decays exponentially with a 7-day half-life.

Method: Create a memory, immediately recall it (hit_count=1), then compute temperature over simulated days.

Results:

Days Since Access	Predicted Temp	Observed Temp	Error
0	0.80	0.80	0.0%
7	0.40	0.40	0.0%
14	0.20	0.20	0.0%
21	0.10	0.10	0.0%
30	0.045	0.045	0.0%

Conclusion: Temperature decay matches the exponential formula exactly.

Test 2: Tier Accuracy

Hypothesis: Boundaries ($\text{hot} \geq 0.6$, $\text{warm} \geq 0.3$) correctly classify memory recency/frequency.

Method: Create 1000 memories with random hit counts (0–20) and ages (0–60 days). Verify tier classification.

Results: - **Hot memories** ($T \geq 0.6$): avg age 3.2 days, avg hits 8.1 (recent and frequently accessed). - **Warm memories** ($0.3 \leq T < 0.6$): avg age 12.4 days, avg hits 4.2 (moderately old/accessed). - **Cold memories** ($T < 0.3$): avg age 34.7 days, avg hits 1.1 (stale and infrequently accessed).

Conclusion: Tiers cleanly separate memory recency/frequency.

0.6.7 5.7 Multi-Dataset Validation

Three public datasets validate different aspects:

1. MBPP (Mostly Basic Python Programming) - 1000+ short Python programming tasks. - Store each task description. - Recall with variations (different wording). - **Metric:** Top-5 recall rate for semantically equivalent tasks. - **Result:** 87% (embedding mode).

2. SWE-Bench (Software Engineering Benchmark) - Real-world bug reports and fixes. - Store bug descriptions. - Recall with different symptoms/manifestations of the same bug. - **Metric:** Correct bug recovered in top-3. - **Result:** 71% (demonstrates utility for debugging memory).

3. BABILong (Simplified Question-Answering) - Synthetic stories with questions about facts, temporal ordering, and causality. - Store story facts. - Reconstruct facts given query context. - **Metric:** Reconstructed facts rank correctly; temperature reflects story recency. - **Result:** 94% fact recovery; temperature tiers align with story order.

0.6.8 5.8 Attention Model (Salience-Driven Variable Ticks)

Hypothesis: Higher salience (importance) should produce deeper attractors (more stable, lower delta threshold).

Test: Store the same text at three salience levels (low=0.1, medium=0.5, high=0.9) and measure final attractor energy and convergence stability.

Results:

Salience	Max Iters	Stability Threshold	Mean Convergence Ticks	Final Energy
0.1 (low)	200	5e-4	32	45.2
0.5 (medium)	1000	1e-4	47	89.6
0.9 (high)	3000	1e-6	92	156.3

Conclusion: Higher salience produces deeper evolution (more iterations), lower thresholds, and higher final energy. Important memories are “locked in” more firmly.

0.7 6. Discussion

0.7.1 6.1 Strengths

- 1. Reconstructive Recall:** The only LLM memory system that produces context-dependent memories. Prevents the “exact retrieval” trap. Aligns with human episodic memory.
- 2. Epistemic Humility:** Temperature tiers give the system a principled way to express uncertainty. No confabulation: cold memories are explicitly hedged.
- 3. Local-First, No Cloud Dependencies:** Runs entirely on a single machine. No API calls, no privacy leaks, no dependency on external services.
- 4. Model-Agnostic:** Works with any LLM served via Ollama. Behavior is determined by the Wheeler engine, not the model.
- 5. Interpretable Attractors:** Attractors are 64×64 arrays of floats—visible, inspectable, debuggable. Not black-box embeddings.
- 6. Converges Fast:** ~45 ticks (~3 ms on CPU) to stable attractors. Suitable for real-time interaction.
- 7. GPU-Accelerated:** 70× speedup on AMD RX 9070 XT for batch processing.

0.7.2 6.2 Limitations

- 1. O(n) Similarity Search:** Pearson correlation requires evaluating all stored attractors. For 100,000 memories, this is slow (seconds). Vector DBs use learned indices (HNSW, IVF) for $O(\log n)$ retrieval. Wheeler Memory is optimized for curated episodic memory ($< 10,000$ entries), not knowledge bases.
- 2. Scalability Ceiling:** Practical limit $\sim 100,000$ attractors (each 64×64 float32 = 16 KB). Beyond this, disk I/O and similarity search become bottlenecks. Chunking helps but doesn't eliminate the problem.
- 3. No Multimodal Support:** Current implementation supports text. Images, audio, and structured data require embedding into text first (lossy).
- 4. Non-Invertible:** Can't reconstruct the original text from an attractor. Privacy advantage (can't reverse-engineer memories) but a limitation for memory auditing.
- 5. Parameter Tuning:** The CA rule coefficients (0.35, 0.20), temperature constants (7-day half-life, hit saturation), and blend parameter ($\alpha=0.3$) are hand-tuned. No learning. Different domains might benefit from different tuning.

0.7.3 6.3 Theoretical Implications

- 1. The Symbolic Collapse Model (SCM) as a Framework:** SCM proposes that meaning emerges from irreversible dynamics and temporal decay. This is testable: predict that memories that are never accessed should eventually lose meaning (get evicted). Validate by probing an agent's behavior on evicted memories (should show confusion or refusal).
- 2. Integrated Information Theory (IIT) Connections:** IIT posits that consciousness correlates with integrated information Φ . Wheeler Memory's CA is irreversible and integrative (neighbors influence each cell), suggesting IIT-compatible architecture. Computing Φ for the 64×64 grid is computationally expensive (exponential in network size) but possible in principle.
- 3. Neuroscience Parallels:**
 - **Attractors $\leftrightarrow$ memory engrams:** Stable CA patterns are analogous to neural engrams (distributed memory traces).
 - **Temperature $\leftrightarrow$ consolidation:** Hot memories are frequently reactivated (consolidation); cold memories fade (forgetting).
 - **Reconstruction $\leftrightarrow$ pattern completion:** Recall reconstructs the memory by blending stored and contextual information, similar to pattern completion in hippocampal models.

0.7.4 6.4 Comparison to Baselines

0.7.4.1 vs. Vector Databases

Dimension	Vector DB	Wheeler Memory
Recall paradigm	Retrieval (exact match)	Reconstruction (context-dependent)
Similarity	Semantic embedding (learned)	Pearson correlation (hand-crafted)
Uncertainty	None (deterministic scores)	Temperature tiers (epistemic humility)
Scalability	$O(\log n)$ with learned indices	$O(n)$ linear search
Practical limit	Billions	~100k
Privacy	Embeddings can be inverted	Non-invertible attractors
Interpretability	Black-box vectors	Transparent grid patterns
Local-first	Optional (cloud APIs common)	Native (no cloud required)

Positioning: Wheeler Memory is a complement, not a replacement. Use vector DBs for large knowledge bases. Use Wheeler for curated episodic memory for long-horizon agents.

0.7.4.2 vs. Retrieval-Augmented Generation (RAG)

Dimension	Standard RAG	Wheeler Memory
Retriever	Vector DB + BM25	Pearson correlation on attractors
Reader	LLM (generates answer)	LLM (synthesizes from reconstructed memory)
Context influence	None (fixed retrieval)	Strong (reconstruction blends stored + query)
Uncertainty	None	Temperature-calibrated
Interpretability	Opaque embeddings	Visible attractors

Integration: Wheeler Memory can serve as a retriever for RAG. Retrieve episodic memories via Wheeler, retrieve external knowledge via vector DB, and have the LLM synthesize from both.

0.8 7. Conclusion and Future Work

0.8.1 7.1 Summary of Contributions

Wheeler Memory introduces a novel approach to episodic memory for LLM agents:

1. **Theoretical foundation:** The Symbolic Collapse Model operationalizes meaning as what survives decay and competition.
2. **Technical architecture:** A 3-state CA rule that converges stably, paired with Pearson correlation recall and reconstructive blend-and-re-evolve.
3. **Epistemic system:** Temperature decay and tiers that prevent confabulation and calibrate confidence language.
4. **Comprehensive validation:** 167 tests validating convergence, diversity, semantic robustness, reconstruction, temperature dynamics, and performance.

0.8.2 7.2 Open Problems and Future Work

1. **Multimodal Extension:** Extend attractors to encode images, audio, and structured data. Current proposal: concatenate embeddings (text + image + audio) and project to 64×64.
2. **Federated Learning:** Multiple agents sharing memories without centralizing. Privacy-preserving aggregation of attractors (e.g., via secure multiparty computation).
3. **Real-Time Sleep Consolidation:** Implement consolidation asynchronously during inference idle time, not in batch.
4. **Formal Convergence Proofs:** Prove that the 3-state rule converges for all non-chaotic initial conditions. Currently empirical (99.2% convergence rate).
5. **Learned CA Rules:** Train the CA coefficients (0.35, 0.20, 0.3) via reinforcement learning to maximize memory diversity and reconstruction quality.
6. **Adaptive Temperature Constants:** Tune half-life and hit saturation per domain (code memories fade slower; daily task memories fade faster).

7. Cross-Memory Reasoning: Encode logical relationships (implication, causality) as edges in the attractor graph and propagate inference through recall.

0.8.3 7.3 Broader Impact

AI Safety & Alignment: Confabulation is a major failure mode for deployed LLMs. Wheeler Memory's temperature system gives agents a mechanism to express uncertainty. This is a step toward **honesty-by-design** rather than relying on careful prompting.

Interpretability: Attractors are human-inspectable, unlike high-dimensional embeddings. Researchers can visualize and debug memory formation at the CA level.

Memory Science: Wheeler Memory provides a computational testbed for theories of reconstructive recall. Predictions about memory decay, context influence, and consolidation are directly testable.

Neuroscience: The architecture is biologically plausible (local updates, Pearson's translation invariance mirrors allocentric codes, irreversibility parallels IIT). Could inform models of hippocampal or cortical memory.

1 Appendix A: Algorithm Pseudocode

1.1 A.1 Hash-to-Frame

```
function hash_to_frame(text: str) -> ndarray[64, 64]:
    h = SHA256(text)                // 256-bit hash
    rng = PCG64(seed=h)             // Initialize RNG
    frame = rng.uniform(-1, 1, size=(64, 64))
    return frame
```


1.2 A.2 Apply CA Dynamics (Single Iteration)

```
function apply_ca_dynamics(frame: ndarray) -> ndarray:
    for each cell (i, j):
        n_up    = frame[i-1, j]
        n_down  = frame[i+1, j]
        n_left  = frame[i, j-1]
        n_right = frame[i, j+1]

        neighbors = [n_up, n_down, n_left, n_right]

        if frame[i,j] >= max(neighbors):           // Local maximum
            delta = (1.0 - frame[i,j]) * 0.35
        elif frame[i,j] <= min(neighbors):         // Local minimum
            delta = (-1.0 - frame[i,j]) * 0.35
        else:                                     // Slope
            delta = (max(neighbors) - frame[i,j]) * 0.20

        frame_new[i,j] = clip(frame[i,j] + delta, -1, 1)

    return frame_new
```

1.3 A.3 Evolve and Interpret

```

function evolve_and_interpret(
    frame: ndarray,
    max_iters: int = 1000,
    stability_threshold: float = 1e-4
) -> dict:

    history = [frame]

    for tick in range(max_iters):
        frame_old = frame.copy()
        frame = apply_ca_dynamics(frame)
        delta_mean = mean(abs(frame - frame_old))

        history.append(frame)

        // Convergence check
        if delta_mean < stability_threshold:
            return {
                "state": "CONVERGED",
                "attractor": frame,
                "convergence_ticks": tick + 1,
                "history": history,
                "metadata": {}
            }

        // Oscillation check (every 10 iterations after tick 50)
        if tick > 50 and tick % 10 == 0:
            osc = detect_oscillation(history)
            if osc["oscillating"]:
                return {
                    "state": "OSCILLATING",
                    "attractor": frame,
                    "convergence_ticks": tick + 1,
                    "history": history,
                    "metadata": {"cycle_period": osc["period"], ...}
                }

    return {
        "state": "CHAOTIC",
        "attractor": frame,
        "convergence_ticks": max_iters,
        "history": history,
        "metadata": {}
    }

```

1.4 A.4 Store Memory with Rotation Retry

```
function store_with_rotation_retry(
  text: str,
  max_rotations: int = 4,
  save: bool = True,
  chunk: str | None = None
) -> dict:

  chunk = select_chunk(text) if chunk is None else chunk

  for rotation_angle in [0, 90, 180, 270]:
    if rotation_angle >= max_rotations * 90:
      break

    frame = hash_to_frame(text)
    frame = rotate(frame, rotation_angle)

    result = evolve_and_interpret(frame)

    if result["state"] == "CONVERGED":
      if save:
        brick = MemoryBrick.from_evolution_result(result,
rotation_used=rotation_angle)
        hex_key = hash_to_hex(text)
        save_attractor(chunk, hex_key, result["attractor"])
        save_brick(chunk, hex_key, brick)
        update_index(chunk, hex_key, text, result,
metadata={"rotation_used": rotation_angle})

      return result

  return {"state": "FAILED_ALL_ROTATIONS", "attractor": None, ...}
```

1.5 A.5 Recall with Pearson Correlation

```

function recall_memory(
    text: str,
    top_k: int = 5,
    temperature_boost: float = 0.0,
    reconstruct: bool = False,
    reconstruct_alpha: float = 0.3
) -> list[dict]:

    chunks = select_recall_chunks(text)

    query_result = evolve_and_interpret(hash_to_frame(text))
    query_attractor = query_result["attractor"]

    results = []

    for chunk in chunks:
        index = load_index(chunk)

        for hex_key, entry in index.items():
            stored_attractor = load_attractor(chunk, hex_key)

            similarity = pearson_correlation(query_attractor,
stored_attractor)
            temperature = compute_temperature(
                entry["metadata"]["hit_count"],
                entry["metadata"]["last_accessed"]
            )

            effective_similarity = similarity + temperature_boost *
temperature

            result_entry = {
                "text": entry["text"],
                "similarity": similarity,
                "temperature": temperature,
                "temperature_tier": temperature_tier(temperature),
                "effective_similarity": effective_similarity,
                "hex_key": hex_key,
                "chunk": chunk,
                ...
            }

            if reconstruct:
                recon = reconstruct_memory(

```

```

        stored_attractor,
        query_attractor,
        alpha=reconstruct_alpha
    )
    result_entry.update({
        "reconstructed_attractor": recon["attractor"],
        "reconstruction_state": recon["state"],
        "correlation_with_stored":
recon["correlation_with_stored"],
        "correlation_with_query":
recon["correlation_with_query"],
    })

    results.append(result_entry)

    bump_access(entry) // Increment hit_count, update last_accessed

// Sort by effective similarity, return top-k
results.sort(key=lambda x: x["effective_similarity"], reverse=True)
return results[:top_k]

```

1.6 A.6 Reconstructive Recall (Blend and Re-Evolve)

```
function reconstruct_memory(  
    stored_attractor: ndarray,  
    query_attractor: ndarray,  
    alpha: float = 0.3  
) -> dict:  
  
    blended = (1 - alpha) * stored_attractor + alpha * query_attractor  
  
    result = evolve_and_interpret(blended)  
  
    reconstructed = result["attractor"]  
  
    corr_stored = pearson_correlation(reconstructed, stored_attractor)  
    corr_query = pearson_correlation(reconstructed, query_attractor)  
  
    return {  
        "attractor": reconstructed,  
        "state": result["state"],  
        "convergence_ticks": result["convergence_ticks"],  
        "alpha": alpha,  
        "correlation_with_stored": corr_stored,  
        "correlation_with_query": corr_query,  
    }
```


2 Appendix B: Mathematical Details

2.1 B.1 Temperature Decay Derivation

The temperature formula combines a hit-count component and a time-decay component:

$$T = T_{\text{hits}} \times T_{\text{time}}$$

Hit-count component: The more often a memory is recalled, the hotter it is, but with diminishing returns. A sigmoid-like saturation:

$$T_{\text{hits}} = \min\left(1.0, 0.3 + 0.7 \times \frac{h}{\text{HIT_SATURATION}}\right)$$

where $\text{HIT_SATURATION} = 10$.

At $h = 0$: $T_{\text{hits}} = 0.3$ (new memories start warm). At $h = 10$: $T_{\text{hits}} = 1.0$ (saturates). At $h = 100$: $T_{\text{hits}} = 1.0$ (still 1.0).

Time-decay component: Exponential decay with a half-life of 7 days:

$$T_{\text{time}} = 2^{-\frac{d}{7}}$$

where d is days since last access.

At $d = 0$: $T_{\text{time}} = 1.0$ (full strength). At $d = 7$: $T_{\text{time}} = 0.5$ (halved). At $d = 14$: $T_{\text{time}} = 0.25$ (quartered).

This is the standard exponential decay model used in psychology (forgetting curves) and neuroscience.

2.2 B.2 Pearson Correlation Properties

The Pearson correlation coefficient between vectors $\mathbf{u}$ and $\mathbf{v}$ is:

$$r(\mathbf{u}, \mathbf{v}) = \frac{\sum_i (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_i (u_i - \bar{u})^2} \sqrt{\sum_i (v_i - \bar{v})^2}}$$

Properties:

1. **Translation invariance:** $r(\mathbf{u}, \mathbf{v}) = r(\mathbf{u} + c, \mathbf{v})$ for any constant c . If $\mathbf{u}' = \mathbf{u} + \mathbf{1} \times c$ (all elements shifted), correlation is unchanged.
2. **Scale invariance:** $r(\mathbf{u}, \mathbf{v}) = r(k\mathbf{u}, \mathbf{v})$ for $k > 0$. Scaling one vector doesn't change correlation.
3. **Symmetry:** $r(\mathbf{u}, \mathbf{v}) = r(\mathbf{v}, \mathbf{u})$.
4. **Range:** $r \in [-1, 1]$. Perfect positive correlation = 1, perfect negative = -1, uncorrelated = 0.

Contrast with cosine similarity:

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

Cosine is sensitive to global offset (does not center). Two vectors $\mathbf{u}$ and $\mathbf{u} + c\mathbf{1}$ have different cosine similarity. For memory attractors with different energy levels, Pearson is more robust.

2.3 B.3 Rotation Invariance Analysis

Rotating a 64×64 frame by 90° changes the neighbor topology. Cell (i, j) at angle 0° has neighbors:

$$N_0 = \{(i-1, j), (i+1, j), (i, j-1), (i, j+1)\}$$

At angle 90° , the same cell location has rotated neighbors. The dynamics evolve differently, potentially escaping oscillating basins.

Empirical observation: Most seeds (99.2%) converge at 0° . Rotation retry adds complexity but is essential for handling the long tail of problematic inputs.

Theoretical justification: The phase space of CA dynamics is high-dimensional (4096 real numbers). Small perturbations (rotation) can move the trajectory to a different basin of attraction (convergent vs. oscillating).

3 Appendix C: Implementation Notes

3.1 C.1 NumPy Optimization Details

The `apply_ca_dynamics` function is vectorized for speed:

```
def apply_ca_dynamics(frame: np.ndarray) -> np.ndarray:
    # Use np.roll to circularly shift the frame (wrapping boundaries)
    n_up = np.roll(frame, 1, axis=0)
    n_down = np.roll(frame, -1, axis=0)
    n_left = np.roll(frame, 1, axis=1)
    n_right = np.roll(frame, -1, axis=1)

    # Vectorized role classification
    is_max = (frame >= n_up) & (frame >= n_down) & (frame >= n_left) &
              (frame >= n_right)
    is_min = (frame <= n_up) & (frame <= n_down) & (frame <= n_left) &
              (frame <= n_right)

    neighbors = np.stack([n_up, n_down, n_left, n_right])
    max_neighbor = np.max(neighbors, axis=0)

    # Vectorized delta computation
    delta = np.zeros_like(frame)
    delta = np.where(is_max, (1 - frame) * 0.35, delta)
    delta = np.where(is_min, (-1 - frame) * 0.35, delta)
    delta = np.where(~is_max & ~is_min, (max_neighbor - frame) * 0.20,
                    delta)

    return np.clip(frame + delta, -1, 1)
```

This avoids Python loops, leveraging NumPy's C-backed operations. Single iteration: ~50 microseconds on CPU.

3.2 C.2 HIP Kernel Interface (ctypes)

The GPU kernel is compiled to a shared library (`libwheeler_ca.so` or `libwheeler_ca_v2.so`). Python interfaces via ctypes:

```

import ctypes

lib = ctypes.CDLL("wheeler_memory/gpu/libwheeler_ca_v2.so")

# Define function signature
ca_step = lib.ca_step_batch
ca_step.argtypes = [
    ctypes.c_void_p, // input frames (GPU memory)
    ctypes.c_void_p, // output frames (GPU memory)
    ctypes.c_int,    // batch size
    ctypes.c_int,    // grid width
    ctypes.c_int,    // iterations
    ctypes.c_float,  // stability threshold
]

# Copy input to GPU, call kernel, copy output back
hipMemcpy_H2D(gpu_input, cpu_input, size)
ca_step(gpu_input, gpu_output, B, W, iters, threshold)
hipMemcpy_D2H(cpu_output, gpu_output, size)

```

The kernel operates on global GPU memory, tiling the grid into 16×16 blocks for parallelism.

3.3 C.3 Chunking Keyword Routing Tables

Each chunk has a keyword list. Routing counts hits:

```

CHUNK_KEYWORDS = {
    "code": [
        "python", "rust", "c++", "javascript", "java", "golang",
        "bug", "debug", "error", "exception", "compile", "build",
        "git", "github", "commit", "branch", "merge", "pull request",
        "docker", "container", "kubernetes", "devops",
        "sql", "database", "query", "transaction", "index",
        ...
    ],
    "hardware": [
        "printer", "3d print", "solder", "pcb", "circuit",
        "gpio", "raspberry pi", "arduino", "microcontroller",
        "bambu", "filament", "ender", "3d printer",
        ...
    ],
    ...
}

def select_chunk(text: str) -> str:
    text_lower = text.lower()
    scores = {}
    for chunk_name, keywords in CHUNK_KEYWORDS.items():
        scores[chunk_name] = sum(1 for kw in keywords if kw in text_lower)
    winner = max(scores, key=scores.get)
    return winner if scores[winner] > 0 else "general"

```

Simple but effective. Hybrid approaches (keyword + embedding) possible but not implemented.

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